CONTENTS		
Chapter no.	**CHAPTER NAME**	**Page no.**
3.	**Vector Calculus**	**3**
	3.1 Introduction to Vector, dot, cross product	
	3.2 Unit vectors, Normals and bases	
	3.3 Vector equations of straight lines and planes	
	3.4 Divergence, Gradient, Curl & the Laplacian	
	3.4.1 Divergence	
	3.4.2 Gradient	
	3.4.3 Curl	
	3.4.4 The Laplacian	
	3.5 Change of coordinates	
	3.5.1 Streamlines for Fluid Flow	
	3.6 Some Vector Identities	
	3.7 Line, Surface and Volume Integral	
	3.7.1 Line Integral	
	3.7.2 Surface Integral	
	3.7.3 Volume Integral	
	3.8 Stokes Theorem	
	3.9 Divergence Theorem	
	3.10 Green's Theorem	
	SET-7	

Career Avenues

Chapter - 3 Vector Calculus

3.1 Introduction to Vector, dot, cross product:

1. Vectors: A vector has a direction, and a length. It is represented by a directed line segment. In a coordinate system it's expressed by components; in space.

The **displacement vector** of a particle can be represented in Cartesian components by

$$\vec{\mathbf{r}}\ t\ = \left\langle x\ t\ , y\ t\ , z\ t\ \right\rangle \rightarrow$$

where t is a **parameter** (time or angle or distance along a curve, etc.)

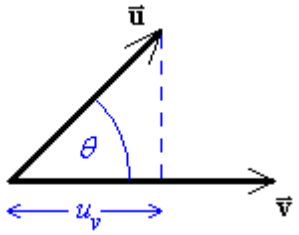
The **distance** of the particle from the origin at any value of t is given by the scalar function

$$r\ t\ = \left|\vec{\mathbf{r}}\ t\ \right| = \sqrt{x\ t\ ^2 + y\ t\ ^2 + z\ t\ ^2}$$

Note:- *the various alternative conventions for a vector and its magnitude:*

$$\mathbf{r} \equiv \vec{\mathbf{r}} \equiv \underline{\mathbf{r}} \qquad \text{and} \qquad r \equiv \left\|\vec{\mathbf{r}}\right\| \equiv \left|\vec{\mathbf{r}}\right|$$

2. Scalar product (dot product):



$$\vec{\mathbf{u}} \bullet \vec{\mathbf{v}}\ = uv\cos\theta$$

The component of vector $\vec{\mathbf{u}}$ in the direction of vector $\vec{\mathbf{v}}$ is

$$u_v\ = \vec{\mathbf{u}} \bullet \hat{\mathbf{v}}\ = u\cos\theta$$

$\vec{\mathbf{u}} \bullet \vec{\mathbf{v}} = 0 \quad \Rightarrow \quad \vec{\mathbf{u}} = \vec{\mathbf{0}}$ or $\vec{\mathbf{v}} = \vec{\mathbf{0}}$ or $\vec{\mathbf{u}} \perp \vec{\mathbf{v}}$ ("$\vec{\mathbf{u}}$ at right angles to $\vec{\mathbf{v}}$")

The scalar product is commutative:

$$\vec{\mathbf{v}} \bullet \vec{\mathbf{u}}\ = \vec{\mathbf{u}} \bullet \vec{\mathbf{v}} \qquad \forall \vec{\mathbf{u}}, \vec{\mathbf{v}}$$

Application. 1) computing lengths and angles: $\cos\theta = \dfrac{\vec{A} \cdot \vec{B}}{\left|\vec{A}\right|\left|\vec{B}\right|}$.

Example: triangle is space with vertices P = (1,0,0), Q = (0,1,0), R = (0,0,2), find angle at P:

$$\cos\theta = \frac{\overrightarrow{PQ} \cdot \overrightarrow{PR}}{\left|\overrightarrow{PQ}\right|\left|\overrightarrow{PR}\right|} = \frac{-1,1,0 \cdot -1,0,2}{\sqrt{2}\sqrt{5}} = \frac{1}{\sqrt{10}}, \qquad \theta \approx 71.5^O$$

Note the sign of dot product: positive if angle less than 90°, negative if angle more than90^O zero if perpendicular.

Career Avenues

Determinant. Definition: det $\vec{A},\vec{B} = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1$

Geometrically: $\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = \pm$ of parallelogram

The sign of 2D determinant has to do with whether $\vec{B}$ is counter clockwise or clockwise from $\vec{A}$ without details

Determinant in space: det $\vec{A},\vec{B},\vec{C} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} a_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$

Geometrically: det $\vec{A},\vec{B},\vec{C} = \pm$ volume of parallelepiped

3. <u>Vector product</u> (cross product):-

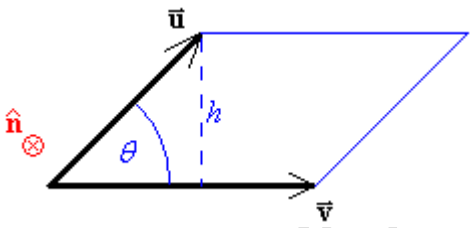
Cross-product. (only for 2 vectors in space): gives a vector, not a scalar (unlike dot-product).

Definition: $\vec{A}\times\vec{B} = \begin{vmatrix} i & j & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = i \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{J} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$

(the 3 x 3 determinant is a symbolic notation, the actual formula is the expansion)

Geometrically: $|\vec{A}\times\vec{B}|$ = area of space parallelogram with sides $\vec{A},\vec{B}$; direction = normal to the plane containing $\vec{A}$ and $\vec{B}$

How to decide between the two perpendicular directions = right – hand rule. 1) extend right hand in direction of $\vec{A}$; 2) curl fingers towards direction of $\vec{B}$; 3) thumb points in same direction as $\vec{A}\times\vec{B}$.



The vector product of two vectors $\mathbf{u}$, $\mathbf{v}$ is

in a direction $\hat{\mathbf{n}}$ at right angles to both $\mathbf{u}$ and $\mathbf{v}$.

$\mathbf{u}\times\mathbf{v} = uv\sin\theta\ \hat{\mathbf{n}}$

The area of the parallelogram is

$$A = |\mathbf{u}\times\mathbf{v}| \quad \text{and} \quad h = u\sin\theta = |\mathbf{u}\times\hat{\mathbf{v}}|$$

The orientation of the plane containing vectors $\mathbf{u}$ and $\mathbf{v}$ is defined by the direction of the vector product $\mathbf{n} = \mathbf{u}\times\mathbf{v}$.

$\mathbf{u}\times\mathbf{v} = \vec{0} \Rightarrow \mathbf{u} = \vec{0}$ or $\mathbf{v} = \vec{0}$ or $\mathbf{u} \| \mathbf{v}$ ("$\mathbf{u}$ parallel to $\mathbf{v}$")

The vector product is anti-commutative:

$\mathbf{v}\times\mathbf{u} \equiv -\mathbf{u}\times\mathbf{v} \Rightarrow \mathbf{v}\times\mathbf{v} \equiv \vec{0}$

Career Avenues

The product rule of differentiation is valid for scalar and vector products:

$$\frac{d}{dt}\,\vec{\mathbf{u}}\bullet\vec{\mathbf{v}} \;=\; \frac{d\vec{\mathbf{u}}}{dt}\bullet\vec{\mathbf{v}} \;+\; \vec{\mathbf{u}}\bullet\frac{d\vec{\mathbf{v}}}{dt} \qquad \text{and} \quad \frac{d}{dt}\,\vec{\mathbf{u}}\times\vec{\mathbf{v}} \;=\; \frac{d\vec{\mathbf{u}}}{dt}\times\vec{\mathbf{v}} \;+\; \vec{\mathbf{u}}\times\frac{d\vec{\mathbf{v}}}{dt} \;=\; -\vec{\mathbf{v}}\times\frac{d\vec{\mathbf{u}}}{dt} \;+\; \vec{\mathbf{u}}\times\frac{d\vec{\mathbf{v}}}{dt}$$

The orientation of the plane containing vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ is defined by the direction of the vector product $\vec{\mathbf{n}} \;=\; \vec{\mathbf{u}}\times\vec{\mathbf{v}}$.

$\vec{\mathbf{u}}\times\vec{\mathbf{v}} = \vec{\mathbf{0}} \;\Rightarrow\; \vec{\mathbf{u}} = \vec{\mathbf{0}}$ or $\vec{\mathbf{v}} = \vec{\mathbf{0}}$ or $\vec{\mathbf{u}} \,\|\, \vec{\mathbf{v}}$ ("$\vec{\mathbf{u}}$ parallel to $\vec{\mathbf{v}}$")

Example 1:-
For the vectors $\vec{\mathbf{u}} \;=\; \langle 4, 3, 2\rangle$ and $\vec{\mathbf{v}} \;=\; \langle 2, -1, -2\rangle$
find $\vec{\mathbf{u}}\bullet\vec{\mathbf{v}}$, u_v, $\vec{\mathbf{u}}\times\vec{\mathbf{v}}$, θ (= the angle between $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$)
and the equation of the plane parallel to $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ that passes through the origin.

Recap of above topics:-

- Vectors are directed line segments - they can be represented in component form or by direction and magnitude. We can use trigonometry and Pythagoras' theorem to switch between the forms
- If $\underline{a} = (x, y, z)$ then $|\underline{a}| = \sqrt{x^2 + y^2 + z^2}$.
- If $\underline{a} = (x, y, z)$ and $\underline{b} = (\alpha, \beta, \gamma)$ then $\underline{a} \pm \underline{b} = (x \pm \alpha, y \pm \beta, z \pm \gamma)$
- If $\underline{a} = (x, y, z)$ then $K\underline{a} = (Kx, Ky, Kz)$ for any value K
- The scalar product of two vectors is used to find the angle between them and the orthogonal projection of one onto the other. The formula is: $\underline{a}\cdot\underline{b} = |\underline{a}||\underline{b}|\cos\theta$ where θ is the angle between the vectors and $\underline{a}\cdot\underline{b}$ means multiply each pair of corresponding coordinates together and then add the results.
- The cross product of two vectors is used to find a vector that is perpendicular to both vectors. The formula is : $\underline{a}\times\underline{b} = \begin{vmatrix} \hat{\underline{i}} & \hat{\underline{j}} & \hat{\underline{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\hat{\underline{i}} - (a_1b_3 - a_3b_1)\hat{\underline{j}} + (a_1b_2 - a_2b_1)\hat{\underline{k}}$ It can be checked that the vector is indeed perpendicular by performing the scalar product with the new vector and each of the original vectors.

3.2 Unit vectors, Normals and bases:

1. Unit Vectors:

A unit vector is a vector whose length is 1. A unit vector is usually denoted by the addition of a 'hat' symbol, $\hat{\underline{a}}$. It is possible to make any vector (except the zero vector, $\underline{0}$) into a unit vector. The formula for this transformation is:

$$\hat{\underline{a}} = \frac{\underline{a}}{|\underline{a}|}$$

The method is to find the modulus of the vector and then to divide the vector by the modulus.

Example:-
$\underline{a} = (2,3,5)$. $|\underline{a}| = \sqrt{2^2 + 3^2 + 5^2} = \sqrt{38}$
$\hat{\underline{a}} = \frac{1}{\sqrt{38}}(2,3,5)$ or, alternatively $\hat{\underline{a}} = (\frac{2}{\sqrt{38}}, \frac{3}{\sqrt{38}}, \frac{5}{\sqrt{38}})$.
It is normally best to keep the square root sign (the surd), rather than use a decimal approximation, in order to preserve accuracy.

Career Avenues

Exercise 1:-
Convert the following vectors into unit vectors:
a) $\underline{a} = (1, 6, 2)$ b) $\underline{b} = (3, -2, 1)$ c) $\underline{c} = (5, 9, -3)$

2. Normal vectors:-

A normal vector is a special type of vector. It is a vector which is perpendicular to a given line or plane.

Normal vectors are used in the process of determining the equation of a plane and are usually denoted $\underline{n}$.

2D Normals:

If you are working in two dimensions then you can find the normal to a line by using one vector on the line.

The simplest way to produce this normal is to swap over the $\hat{\underline{i}}$ and $\hat{\underline{j}}$ components and then change the sign on one component.

For example:

Given $\underline{a} = (2, 3)$ a normal to the line containing $\underline{a}$ is $\underline{n} = (-3, 2)$. Equivalently, $\underline{n} = (3, -2)$ is also a normal to this line.

Each line has two normals. These normals face in opposite directions. It is usually unimportant which one you use.

In order to check that you have found a normal, you can use the dot product with the vector on the line; the result should be zero.

For example:

Is $(7, -5)$ normal to the line containing $(5, 7)$? $(5, 7)\cdot(7, -5) = 35 + -35 = 0$. The vectors are perpendicular, so $(7, -5)$ is normal to the given line.

3D Normals:-

In three dimensions we look for vectors which are normal to a given plane. To do this we need to find a vector perpendicular to two vectors which lie within the plane.

The usual method for this is to use the cross product.

For example:- to find the normal to the plane containing $\underline{a} = (2, -1, 3)$ and $\underline{b} = (-2, 4, 1)$ you would work out:

$$\underline{a}\times\underline{b} = \begin{vmatrix} \hat{\underline{i}} & \hat{\underline{j}} & \hat{\underline{k}} \\ 2 & -1 & 3 \\ -2 & 4 & 1 \end{vmatrix} = ((-1)\times 1 - 3\times 4)\hat{\underline{i}} - (2\times 1 - 3\times(-2))\hat{\underline{j}} + (2\times 4 - (-1)\times(-2))\hat{\underline{k}}$$

$$= (-1-12)\hat{\underline{i}} - (2+6)\hat{\underline{j}} + (8-2)\hat{\underline{k}}$$

$$= -13\hat{\underline{i}} - 8\hat{\underline{j}} + 6\hat{\underline{k}}$$

So the normal to the plane is $\underline{n} = (-13, -8, 6)$ which can be checked via the scalar product as before.

Once you have found one normal, you can find the normal that goes in the opposite direction by multiplying it by -1.

Career Avenues

Career Avenues

Exercise 2:-

1. Find a normal to the line containing:

a) (6, 8) b) (23, -2) c) (197, 356)

2. Find a normal to the plane containing the following points:

a) $\underline{a} = (6, 3, 1), \quad \underline{b} = (0, 2, 5)$

b) $\underline{c} = (3, 1, 2), \quad \underline{d} = (4, 0, 3)$

c) $\underline{e} = (-6, 2, 4), \quad \underline{f} = (1, 7, 2)$

3. Bases:

A basis is a set of vectors which can be used to produce every other vector. A basis is said to 'span' a space. The vectors in a basis must be linearly independent, i.e. there is no way of adding multiples of the vectors together to produce the zero vector, other than when you multiply them all by 0.

If the vectors which form a basis are orthogonal to each other, i.e. at right angles to each other, then they are said to form an orthogonal basis.The basis which is used most frequently is the $\hat{\underline{i}}, \hat{\underline{j}}, \hat{\underline{k}}$ basis, where

$\hat{\underline{i}} = 1,0,0$, $\hat{\underline{j}} = 0,1,0$, $\hat{\underline{k}} = 0,0,1$.

In 2 dimensions this is just $\hat{\underline{i}} = 1,0$, $\hat{\underline{j}} = 0,1$

(Note that the hat symbol is not always used for $\underline{i}, \underline{j}, \underline{k}$)

Any vector in 3-dimensional space can be written as the sum of multiples of the 3 basis vectors.

Example:-

$(2,6,-4) = 2(1,0,0) + 6(0,1,0) - 4(0,0,1) = 2\hat{\underline{i}} + 6\hat{\underline{j}} - 4\hat{\underline{k}}$

There are other possible bases for 3 dimensional space, but $\hat{\underline{i}}, \hat{\underline{j}}, \hat{\underline{k}}$ is the simplest to use.

To express a vector in terms of another basis, you need to use simultaneous equations in order to find out the multiples required.

Example:-

Express the vector $(1,3,-6)$ in terms of the basis vectors $(1,1,0), (1,0,1), (0,1,1)$.

Using $\alpha, \beta,$ and γ to represent the unknown multiples we have:

$(1,3,-6) = \alpha(1,1,0) + \beta(1,0,1) + \gamma(0,1,1)$

Splitting this by coordinates we get:

1 $1 = \alpha + \beta$, **2** $3 = \alpha + \gamma$, **3** $-6 = \beta + \gamma$

Rearranging 1. gives $\alpha = 1 - \beta$, so substituting the value of α into **2** gives $3 = 1 - \beta + \gamma \Rightarrow 2 = \gamma - \beta$.

Adding this to the last equation gives $-4 = 2\gamma \Rightarrow \gamma = -2$. By substituting the value for γ back in to the other equations, we find that $\alpha = 5$ and $\beta = -4$.

Hence $(1,3,-6) = 5(1,1,0) - 4(1,0,1) - 2(0,1,1)$

3.3 Vector equations of straight lines and planes:

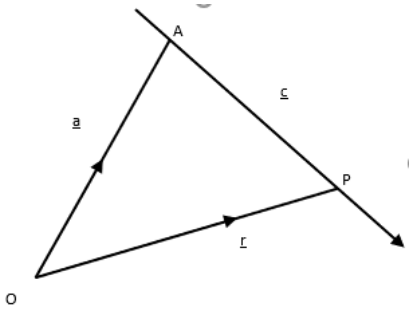
Vectors can be used to find the equation of a line. There are two different methods which are used depending on what information is available.

1. Equation of a line given direction:

Career Avenues

This requires the following information: A vector which lies on the line, and a vector which is in the direction of the line.

If a lie on the line which is in the direction of c, then choose an arbitrary point P on the line, with position vector r from an origin O. This produces the diagram:



To find the equation for the line we need to express $\overrightarrow{AP}$ in terms of the other vectors.

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \underline{r} - \underline{a}$

However as $\overrightarrow{AP}$ is a part of the vector c, it can also be expressed as $\overrightarrow{AP} = \lambda \underline{c}$ where λ is a scalar.

Hence $\underline{r} - \underline{a} = \lambda \underline{c}$ which rearranges to $\underline{r} = \underline{a} + \lambda \underline{c}$, the equation of the line.

Example:-

Find the equation of the line through $\underline{a} = 2,3$ and in the direction of $\underline{c} = (1,-2)$.

Using the formula, $\underline{r} = \underline{a} + \lambda \underline{c}$, gives $\underline{r} = (2,3) + \lambda(1,-2)$.

This can be simplified to $\underline{r} = (2+\lambda,\ 3-2\lambda)$.

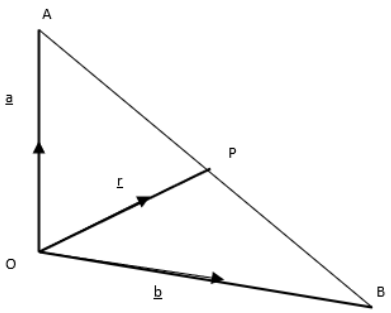
Hence some points on this line (found by changing the value of λ) are: (2,3), (3,1), (4,-1).

Exercise 4:-

1. Find the equation of the line through the point $\underline{a} = (2,6)$ and in the direction of $\underline{c} = (5,2)$
2. Find the equation of the line through the point $\underline{a} = (3,7)$ and in the direction of $\underline{c} = (-1,6)$

2. Equation of a line given 2 points:-

This requires two fixed points on the line, A and B with position vectors a and b respectively and P a general point on the line with position vector r. Here the diagram is:



Career Avenues

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \underline{r} - \underline{a}$

But $\overrightarrow{AP} = \lambda(\overrightarrow{AB})$ and $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \underline{b} - \underline{a}$.

Hence $\underline{r} - \underline{a} = \lambda(\underline{b} - \underline{a})$.

This can be rearranged into $\underline{r} = \underline{a} + \lambda(\underline{b} - \underline{a})$.

Note that if $\lambda = 0$, then $\underline{r} = \underline{a}$ and if $\lambda = 1$, then $\underline{r} = \underline{b}$

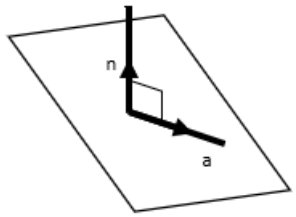
Example :-
Find the equation of the line which passes through the points $\underline{a} = (2,5),\ \underline{b} = (4,9)$.
Substitute the points into the formula $\underline{r} = \underline{a} + \lambda(\underline{b} - \underline{a})$:
$\underline{r} = (2,5) + \lambda((4,9) - (2,5))$ then perform the subtraction:
$\underline{r} = (2,5) + \lambda(2,4) = (2 + 2\lambda, 5 + 4\lambda)$

Exercise 5:-
1. Find the equation of the line which passes through the points $\underline{a} = (4,-2),\ \underline{b} = (3,9)$
2. Find the equation of the line which passes through the points $\underline{a} = (1,6),\ \underline{b} = (8,-3)$

3. Equation of a plane:

The equation of a plane requires the use of a normal vector.

Given the position vector $\underline{a}$ of a point on the plane and a vector $\underline{n}$ which is normal to the plane, the equation of a plane is:



$\underline{r} \cdot \underline{n} = \underline{a} \cdot \underline{n}$

where $\underline{r} = (x, y, z)$. i.e. $\underline{r}$ is a general (unknown) point on the plane and the dot is the scalar product.

In order to obtain the vector equation of a plane, work out the right hand side (RHS) and leave the left alone. In order to obtain the Cartesian equation of a plane, work out both sides.

Example:-
Find the equation of the plane through $\underline{a} = (1,7,3)$ with normal $\underline{n} = (2,-8,1)$.
First, substitute the values into the formula:
$\underline{r} \cdot (2,-8,1) = (1,7,3) \cdot (2,-8,1)$
To get the vector equation, perform the dot product on the RHS:
$\underline{r} \cdot (2,-8,1) = -51$.
To get the Cartesian form, also work out the LHS:
$(x, y, z) \cdot (2,-8,1) = -51 \Rightarrow 2x - 8y + z = -51$

Career Avenues

Exercise 6:-
Find the vector and Cartesian equations of the following planes:
a) Containing the point $\underline{a} = (1,2,5)$ and with normal $\underline{n} = (2,-4,6)$;
b) Containing the point $\underline{a} = (3,-2,4)$ and with normal $\underline{n} = (1,6,2)$

4. Intersections of lines and planes:-

Intersection of two lines:-

Given the equations of 2 lines, $\underline{r_1} = \underline{a} + \lambda\underline{c}$, $\underline{r_2} = \underline{b} + \mu\underline{d}$, the lines intersect if you can find values λ, μ such that $\underline{r_1} = \underline{r_2}$

For example, $\underline{r} = (1,5,4) + \lambda(4,2,3), \quad \underline{r} = (6,7,-1) + \mu(3,2,11)$

Set the equations equal to each other, so

$(1,5,4) + \lambda(4,2,3) = (6,7,-1) + \mu(3,2,11)$

Combine the two vectors not involving λ or μ :

$\lambda(4,2,3) = (5,2,-5) + \mu(3,2,11)$.

Split into components in order to get 3 simultaneous equations:

$4\lambda = 5 + 3\mu, \quad 2\lambda = 2 + 2\mu, \quad 3\lambda = -5 + 11\mu$

Solve to find λ and μ .

(If you need help with this part please refer to Algebra 4.)

In this example $\lambda = 2$ and $\mu = 1$.

To find the point where the lines cross, substitute the value of λ or μ into the equation for one of the lines.

Here, substituting λ into the first equation gives

$(1,5,4) + \lambda(4,2,3) = (1,5,4) + 2(4,2,3)$
$= (1,5,4) + (8,4,6) = (9,9,10)$

You should always check this result with the other equation to ensure it is correct.

5. Intersection of two planes:

Given the normals ($\underline{n_1}$ and $\underline{n_2}$) to the two planes, the line of intersection of the two planes can be found from:

$\underline{r} = \underline{c} + \lambda\underline{n}$

where $\underline{n} = \underline{n_1} \times \underline{n_2}$ and $\underline{c}$ is a point common to both planes.

If you are given the Cartesian forms of the planes, then you can find the normal to the plane by taking the coefficients of $x, y,$ and z . For example, the normal to $x + 3y + 4z = 24$ is $(1,3,4)$.

Example:-
Find the equation of the line of intersection of the planes:
$\underline{r_1} \cdot (2,4,1) = 14, \quad \underline{r_2} \cdot (1,-2,5) = 5$
Here we have $\underline{n_1} = (2,4,1)$ and $\underline{n_2} = (1,-2,5)$

Career Avenues

$$\underline{n}_1 \times \underline{n}_2 = \begin{vmatrix} \underline{\hat{i}} & \underline{\hat{j}} & \underline{\hat{k}} \\ 2 & 4 & 1 \\ 1 & -2 & 5 \end{vmatrix} = (4\times5)-(1\times-2)\ \underline{\hat{i}} - (2\times5)-(1\times1)\ \underline{\hat{j}} + (2\times-2)-(4\times1)\ \underline{\hat{k}}$$

$= 22\underline{\hat{i}} - 9\underline{\hat{j}} - 8\underline{\hat{k}}$ so $\underline{n} = (22,-9,-8)$

Now we need a point common to both planes. The simplest way to do this is to try setting $z = 0$. This means that we only need deal with 2 equations in 2 unknowns.

To find the common point, set $z = 0$ in the Cartesian equations of the planes.

This gives $2x + 4y = 14$ and $x - 2y = 5$. Solving these simultaneous equations gives $x = 6$, $y = \frac{1}{2}$.

Hence $\underline{c} = (6, \frac{1}{2}, 0)$ and the equation of the line of intersection is: $\underline{r} = (6, \frac{1}{2}, 0) + \lambda(22,-9,8)$.

Exercise 7:-

Find the equation of the line of intersection for the planes:

a) $\underline{r}_1 \cdot (6,3,4) = 12, \quad \underline{r}_2 \cdot (-2,3,1) = 4$

b) $2x + 4y - 3z = 14, \ x - 2y + 4z = 3$

3.4 Divergence, Gradient, Curl & the Laplacian:

Divergence, Gradient and Curl are used to give information about a vector field. (A vector field is a field where each point can be described as a vector.) All three of these require knowledge of calculus, namely differentiation and partial differentiation. If you are not familiar with these concepts or need to brush up a little, please look at the booklet Differentiation 1 before continuing.

Divergence, gradient and curl all utilise the symbol ∇ (del or nabla).

∇ represents the vector $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = \frac{\partial}{\partial x}\underline{\hat{i}} + \frac{\partial}{\partial y}\underline{\hat{j}} + \frac{\partial}{\partial z}\underline{\hat{k}}$.

3.4.1 Divergence:

Divergence is used to measure the density of a vector field in order to see how quickly matter is moving into and out of a region.

Divergence is the flux per unit volume at a point:

Div is calculated from a vector and the result is a scalar.

$\nabla \cdot F$ is used to denote divergence.

Divergence is found by taking the dot product of ∇ and the vector.

$$\nabla \cdot F = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Here F_x, F_y, F_z are the components of the vector F.

To calculate Divergence differentiate the first component with respect to x, then differentiate the second component with respect to y and differentiate the third component with respect to z. Finally, add the results of the 3 differentiations together.

Examples:-

Career Avenues

1. If $F = 2x\hat{i} + 3xy\hat{j} + z^2\hat{k}$, then

$$\nabla \cdot F = \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(3xy) + \frac{\partial}{\partial z}(z^2)$$
$$= 2 + 3x + 2z$$

2. If $F = x^2\hat{i} + 2z\hat{j} - 3xy\hat{k}$ then

$$\nabla \cdot F = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(-3xy)$$
$$= 2x + 0 + 0 = 2x$$

Note that the results do not include $\hat{i}$, $\hat{j}$, or $\hat{k}$, they are scalars.

Exercise 8:-

Find the divergence of the following vector fields:

a) $F = 4x^2y\hat{i} + 2yz\hat{j} - 3z\hat{k}$

b) $F = 3y\hat{i} + 2x^2y\hat{j} + 2yz\hat{k}$

c) $F = 3\cos 2x\hat{i} + 3y\sin 2x\hat{j} + 2z\cos x\hat{k}$

d) $F = 3e^x\hat{i} + y^2e^x\hat{j} + e^{2xz}\hat{k}$

3.4.2 Gradient:

The gradient or Grad is used to find where in the field the gradient is changing the most rapidly. The gradient function is used on a scalar and the result is a vector.

$$\nabla f = \frac{\partial}{\partial x}(f)\hat{i} + \frac{\partial}{\partial y}(f)\hat{j} + \frac{\partial}{\partial z}(f)\hat{k}$$

Here f is the scalar that you are differentiating.

To calculate Grad differentiate the whole function with respect to x. This gives the $\hat{i}$ component. Next differentiate the original function with respect to y. This gives the $\hat{j}$ component. Finally repeat for z, giving the $\hat{k}$ component.

Examples:-

1. If $f(x, y, z) = 2z^2 + 3yx$ then

$$\nabla f = \frac{\partial}{\partial x}(2z^2 + 3yx)\hat{i} + \frac{\partial}{\partial y}(2z^2 + 3yx)\hat{j} + \frac{\partial}{\partial z}(2z^2 + 3yx)\hat{k}$$
$$= 3y\hat{i} + 3x\hat{j} + 4z\hat{k}$$

2. If $f(x, y, z) = 5xyz^2 - 2xy + \cos x$ then

$$\nabla f = \frac{\partial}{\partial x}(5xyz^2 - 2xy + \cos x)\hat{i} + \frac{\partial}{\partial y}(5xyz^2 - 2xy + \cos x)\hat{j}$$
$$+ \frac{\partial}{\partial z}(5xyz^2 - 2xy + \cos x)\hat{k}$$
$$= (5yz^2 - 2y - \sin x)\hat{i} + (5xz^2 - 2x)\hat{j} + (10xyz)\hat{k}$$

Exercise 9:-

Find the gradient vector for the following:

Career Avenues

a) $f = x^2y + 3yz + 2xz^2$

b) $f = 3x^2y^2z^3$

c) $f = x\cos y + z^2 \sin x + xy \sin z$

d) $f = x^2e^y + ye^{xz}$

Career Avenues

3.4.3 Curl:

Curl is used to determine the amount of rotation of a vector field. If the curl of a vector field is zero, then it is said to be 'irrotational', i.e. it does not rotate. The curl is used on a vector and the result of calculating the curl is also a vector.

Calculating the curl is very similar to calculating the cross product of two vectors and is denoted $\nabla \times F$.

The main difference is that the first vector that is used here is $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$.

The formula for curl is:

$$\nabla \times F = \begin{vmatrix} \hat{\underline{i}} & \hat{\underline{j}} & \hat{\underline{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial F_z}{\partial y} - \frac{dF_y}{\partial z}\right)\hat{\underline{i}} - \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z}\right)\hat{\underline{j}} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{\underline{k}}$$

To calculate the curl, first write out the determinant with $\hat{\underline{i}}, \hat{\underline{j}}, \hat{\underline{k}}$ as the top row, ∇ as the second row and the vector F as the third. The next step is to work it out as if it were the cross product. The difference here is that instead of finding pairs to multiply together, you are finding out which part of F will be differentiated by x, y, or z.

Curl is the rotation per unit area at a given point.

Examples:-

1. If $F = x^2\hat{\underline{i}} + 2z\hat{\underline{j}} - 3xy\hat{\underline{k}}$ then

$$\nabla \times F = \begin{vmatrix} \hat{\underline{i}} & \hat{\underline{j}} & \hat{\underline{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 2z & -3xy \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial y}(-3xy) - \frac{\partial}{\partial z}(2z)\right)\hat{\underline{i}} - \left(\frac{\partial}{\partial x}(-3xy) - \frac{\partial}{\partial z}(x^2)\right)\hat{\underline{j}} + \left(\frac{\partial}{\partial x}(2z) - \frac{\partial}{\partial y}(x^2)\right)\hat{\underline{k}}$$

$$= -3x - 2\ \hat{\underline{i}} - -3y - 0\ \hat{\underline{j}} + 0 - 0\ \hat{\underline{k}}$$

$$= (-3x - 2)\hat{\underline{i}} + 3y\hat{\underline{j}}$$

2. If $F = 2xy^2z\hat{\underline{i}} + 3xy\hat{\underline{j}} + z^2\hat{\underline{k}}$, then

$$\nabla \times F = \begin{vmatrix} \hat{\underline{i}} & \hat{\underline{j}} & \hat{\underline{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy^2z & 3xy & z^2 \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(3xy)\right)\hat{\underline{i}} - \left(\frac{\partial}{\partial x}(z^2) - \frac{\partial}{\partial z}(2xy^2z)\right)\hat{\underline{j}} + \left(\frac{\partial}{\partial x}(3xy) - \frac{\partial}{\partial y}(2xy^2z)\right)\hat{\underline{k}}$$

$$= 0 - 0\ \hat{i} - 0 - 2xy^2\ \hat{j} + 3y - 4xyz\ \hat{k}$$

$$= 2xy^2\hat{\underline{j}} + (3y - 4xyz)\hat{k}$$

Career Avenues

Career Avenues

Exercise 10:-
Calculate the curl of the following vector fields:

a) $F = x^2 y\hat{\underline{i}} + z^3 y\hat{\underline{j}} + x\hat{\underline{k}}$

b) $F = y^2 z\hat{\underline{i}} + x^3 z^2\hat{\underline{j}} + 3y\hat{\underline{k}}$

c) $F = 3y\cos 2x\hat{\underline{i}} + 3x\sin 2z\hat{\underline{j}} + \cos y\hat{\underline{k}}$

d) $F = ze^{xy}\hat{\underline{i}} + z^2 e^x\hat{\underline{j}} + xe^{3y}\hat{\underline{k}}$

Possible combinations of Divergence, Gradient and Curl
The combinations which work depends on whether the result of the first process produces a vector or a scalar.
As Grad produces a vector, it is possible to find Div (Grad) and Curl (Grad). As Div produces a scalar it is possible to find Grad (Div). As Curl produces a vector, it is possible to find Div(Grad), and as Curl also works on vectors, you can find Curl(Curl) too.

3.4.4 The Laplacian:

The Laplacian is denoted by ∇^2.

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

The Laplacian works on scalars and produces a scalar result.

To find $\nabla^2 f$, find the first derivative of f with respect to x, y, and z. Then find the second derivative of each and, finally add up your results

Example:-
Given, $f = 3x^2 y + xz^2$, find $\nabla^2 f$.

$$\begin{aligned}\nabla^2 f &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \\ &= \frac{\partial^2}{\partial x^2}(3x^2 y + xz^2) + \frac{\partial^2}{\partial y^2}(3x^2 y + xz^2) + \frac{\partial^2}{\partial z^2}(3x^2 y + xz^2) \\ &= \frac{\partial}{\partial x}(6xy + z^2) + \frac{\partial}{\partial y}(3x^2) + \frac{\partial}{\partial z}(2xz) \\ &= 6y + 0 + 2x = 6y + 2x\end{aligned}$$

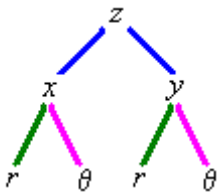
Exercise 11:-
Find the Laplacian of the following functions:
a) $x^2 + 3yz^3$ b) $xy^2 z^4 + 3x^3 y$

Career Avenues

3.5 Change of coordinates:

Suppose z = f (x, y) (where (x, y) are Cartesian coordinates) and $\frac{\partial z}{\partial r}$ is wanted, (where (r, θ) are plane polar coordinates).

Then



$$\Rightarrow \quad \frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r}$$

But x = r cos θ, y = r sin θ

$$\Rightarrow \quad \frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$$

$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} - r\sin\theta + \frac{\partial z}{\partial y}\, r\cos\theta$ can be found in a similar way.

In matrix form, the chain rule can be expressed concisely as

$$\begin{bmatrix} \frac{\partial z}{\partial r} \\ \frac{\partial z}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{bmatrix} \quad .$$

Note that

$$\frac{\partial\ x, y}{\partial\ r, \theta} = \text{abs}\left(\det\begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{bmatrix}\right) \text{ is the } \textbf{Jacobian}$$

For the transformation from Cartesian to plane polar coordinates in ^{2}, the Jacobian is

$$\frac{\partial\ x, y}{\partial\ r, \theta} = \begin{Vmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{Vmatrix} = \left| r\cos^2\theta + r\sin^2\theta \right| = r$$

Integrals over areas can therefore be transformed using the Jacobian:

$$\iint_A f\ x, y\ dx\, dy = \iint f\ x, y\ \frac{\partial\ x, y}{\partial\ r, \theta} dr\, d\theta = \iint_A g\ r, \theta\ \ r\, dr\, d\theta$$

where f (x, y) = g(r, θ) at all points in the area A of integration.

Career Avenues

3.5.1 Streamlines for Fluid Flow:

Let **v**(**r**) be the velocity at any point (x, y, z) in an incompressible fluid. Because the fluid is incompressible, the flow in to any region must be matched by the flow out from that region (except when the region includes a source or a sink). This generates the continuity equation

div **v** = 0

Let us take the case of fluid flow parallel to the x-y plane everywhere, so that we can ignore the third dimension and consider the flow in two dimensions only. Then

$$\vec{\mathbf{v}} = \vec{\mathbf{v}}\ x, y\ = u\ x, y\ \hat{\mathbf{i}} + v\ x, y\ \hat{\mathbf{j}}$$

curl **F = 0** everywhere $\Rightarrow$ **F** is an **irrotational vector field**.

Exercise:
Find curl **F** for **F** = +cos y, −sin x, 0, .
Also find the lines of force for the vector field **F**.

All differentiable gradient-vector fields are irrotational:

$$\text{curl grad}\,\phi \equiv \vec{\nabla}\times\vec{\nabla}\phi \equiv \vec{\mathbf{0}}$$

The **Laplacian** of a twice-differentiable scalar field ϕ is:

$$\text{div grad}\,\phi \equiv \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \bullet \left\langle \frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z} \right\rangle \equiv \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \equiv \nabla^2 \phi$$

Laplace's equation is $\nabla^2 \phi \equiv \dfrac{\partial^2 \phi}{\partial x^2} + \dfrac{\partial^2 \phi}{\partial y^2} + \dfrac{\partial^2 \phi}{\partial z^2} = 0$

3.6 Some Vector Identities:

1. $\vec{\nabla}\times\vec{\nabla} f$ = curl grad f = $\vec{\mathbf{0}}$

2. $\vec{\nabla}\cdot\vec{\nabla}\times\vec{\mathbf{F}}$ = div curl $\vec{\mathbf{F}}$ = 0

Laplacian of V = $\nabla^2 V$ = $\vec{\nabla}\cdot\ \vec{\nabla} V$ = div grad V

3. $\vec{\nabla}^2\vec{\mathbf{F}} = \vec{\nabla}\ \vec{\nabla}\cdot\vec{\mathbf{F}} - \vec{\nabla}\times\ \vec{\nabla}\times\vec{\mathbf{F}}$ = grad div $\vec{\mathbf{F}}$ − curl curl $\vec{\mathbf{F}}$

4. $\vec{\nabla}\ f g = f\ \vec{\nabla} g + \vec{\nabla} f\ g$

5. $\vec{\nabla}\cdot\ g\vec{\mathbf{F}} = \vec{\nabla} g\ \cdot\vec{\mathbf{F}} + g\ \vec{\nabla}\cdot\vec{\mathbf{F}}$

6. div (g **F**) = (grad g)•**F** + g div **F**

7. $\vec{\nabla}\times\ g\vec{\mathbf{F}} = \vec{\nabla} g\ \times\vec{\mathbf{F}} + g\ \vec{\nabla}\times\vec{\mathbf{F}}$

8. curl (g **F**) = (grad g)×**F** + g curl **F**

Career Avenues

9. $\vec{\nabla}\cdot\ \vec{\mathbf{F}}\times\vec{\mathbf{G}}\ =\ \vec{\nabla}\times\vec{\mathbf{F}}\ \cdot\vec{\mathbf{G}}\ -\ \vec{\mathbf{F}}\cdot\ \vec{\nabla}\times\vec{\mathbf{G}}$

10. div (**F** × **G**) = (curl **F**)•**G** − **F**•(curl **G**)

11. $\vec{\nabla}\times\ \vec{\mathbf{F}}\times\vec{\mathbf{G}}\ =\ \vec{\mathbf{G}}\cdot\vec{\nabla}\ \vec{\mathbf{F}}\ -\ \ \vec{\mathbf{F}}\cdot\vec{\nabla}\ \vec{\mathbf{G}}\ +\ \ \vec{\nabla}\cdot\vec{\mathbf{G}}\ \vec{\mathbf{F}}\ -\ \ \vec{\nabla}\cdot\vec{\mathbf{F}}\ \vec{\mathbf{G}}$

where $\vec{\mathbf{F}}\cdot\vec{\nabla}\ \vec{\mathbf{G}}\ =\ \left(F_1\frac{\partial G_1}{\partial x}\ +\ F_2\frac{\partial G_1}{\partial y}\ +\ F_3\frac{\partial G_1}{\partial z}\right)\hat{\mathbf{i}}$

$+\ \left(F_1\frac{\partial G_2}{\partial x}\ +\ F_2\frac{\partial G_2}{\partial y}\ +\ F_3\frac{\partial G_2}{\partial z}\right)\hat{\mathbf{j}}$

$+\ \left(F_1\frac{\partial G_3}{\partial x}\ +\ F_2\frac{\partial G_3}{\partial y}\ +\ F_3\frac{\partial G_3}{\partial z}\right)\hat{\mathbf{k}}$

so that $\vec{\mathbf{F}}\cdot\vec{\nabla}$ is the operator $\left(F_1\frac{\partial}{\partial x}\ +\ F_2\frac{\partial}{\partial y}\ +\ F_3\frac{\partial}{\partial z}\right)$

12. $\vec{\nabla}\ \vec{\mathbf{F}}\cdot\vec{\mathbf{G}}\ =\ \vec{\mathbf{G}}\cdot\vec{\nabla}\ \vec{\mathbf{F}}\ +\ \ \vec{\mathbf{F}}\cdot\vec{\nabla}\ \vec{\mathbf{G}}\ +\ \vec{\mathbf{G}}\times\ \vec{\nabla}\times\vec{\mathbf{F}}\ \ +\ \vec{\mathbf{F}}\times\ \vec{\nabla}\times\vec{\mathbf{G}}$

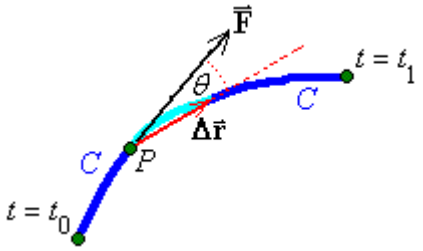
3.7 Line, Surface and Volume Integral:

3.7.1 Line integrals:

Two applications of line integrals are treated here: the evaluation of work done on a particle as it travels along a curve in the presence of a [vector field] force; and the evaluation of the location of the centre of mass of a wire.

Work done:

The work done by a force **F** in moving an elementary distance **Δr** along a curve C is approximately the product of the component of the force in the direction of **Δr** and the distance | **Δr** | travelled:



$\Delta W\ \approx\ \vec{\mathbf{F}}\bullet\Delta\vec{\mathbf{r}}\ =\ F\cos\theta\,|\,\Delta\vec{\mathbf{r}}\,|$

Integrating along the curve C yields the total work done by the force **F** in moving along the curve C:

$$W\ =\ \int_C \vec{\mathbf{F}}\bullet\mathbf{d}\vec{\mathbf{r}}$$

$$=\ \int_C f_1\,dx+f_2\,dy+f_3\,dz\ =\ \int_{t_0}^{t_1}\left(f_1\frac{dx}{dt}+f_2\frac{dy}{dt}+f_3\frac{dz}{dt}\right)dt$$

$$\therefore\ \ W\ =\ \int_C \vec{\mathbf{F}}\bullet\mathbf{d}\vec{\mathbf{r}}\ =\ \int_C \vec{\mathbf{F}}\bullet\frac{d\vec{\mathbf{r}}}{dt}\,dt$$

Career Avenues

Example:

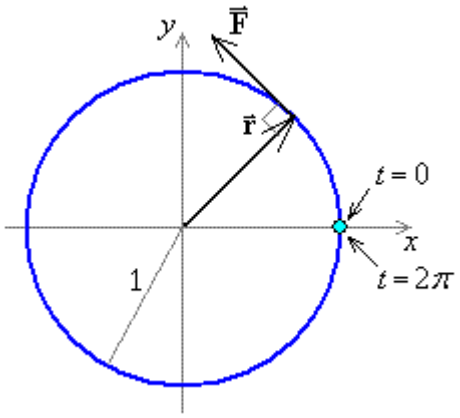
Find the work done by $\vec{\mathbf{F}} = \langle -y, x, z \rangle$ in moving around the curve C (defined in parametric form by x = cos t , y = sin t , z = 0 , $0 \le t \le 2\pi$).

$\vec{\mathbf{F}} = \langle -y, x, z \rangle\big|_C = \langle -\sin t, \cos t, 0 \rangle$

$\dfrac{d\vec{\mathbf{r}}}{dt} = \dfrac{d}{dt}\langle \cos t, \sin t, 0 \rangle = \langle -\sin t, \cos t, 0 \rangle$

$\Rightarrow \quad \vec{\mathbf{F}} \bullet \dfrac{d\vec{\mathbf{r}}}{dt} = \sin^2 t + \cos^2 t + 0 = 1$

$\Rightarrow \quad W = \displaystyle\int_0^{2\pi} \vec{\mathbf{F}} \bullet \frac{d\vec{\mathbf{r}}}{dt}\, dt = \int_0^{2\pi} 1\, dt = \underline{\underline{2\pi}}$



Note that $F_v = \vec{\mathbf{F}} \bullet \hat{\mathbf{v}} = \dfrac{\vec{\mathbf{F}} \bullet \dfrac{d\vec{\mathbf{r}}}{dt}}{\left| \dfrac{d\vec{\mathbf{r}}}{dt} \right|} = 1$ everywhere on the curve C, so that

$W = 1 \times C = 2\pi$ (the length of the path around the circle).

Also note that $\vec{\mathbf{F}} = \langle -y, x, z \rangle \quad \Rightarrow \quad \text{curl}\, \vec{\mathbf{F}} = 2\hat{\mathbf{k}}$ everywhere in $\mathbb{R}^3$.

The lesser curvature of the circular lines of force further away from the z axis is balanced exactly by the increased transverse force, so that curl **F** is the same in all of $\mathbb{R}^3$.

We shall see later (in Stokes' theorem) that the work done is also the normal component of the curl integrated over the area enclosed by the closed curve C. In this case

$W = \vec{\nabla} \times \vec{\mathbf{F}} \bullet \hat{\mathbf{n}}\, A = 2\hat{\mathbf{k}} \bullet \hat{\mathbf{k}}\, \pi\, 1^2 = 2\pi$.

Example:

Find the work done by $\vec{\mathbf{F}} = \langle x, y, z \rangle$ in moving around the curve C (defined in parametric form by x = cos t , y = sin t , z = 0 , $0 \le t \le 2\pi$).

$\vec{\mathbf{F}} = \langle x, y, z \rangle\big|_C = \langle \cos t, \sin t, 0 \rangle = \vec{\mathbf{r}}$

$\dfrac{d\vec{\mathbf{r}}}{dt} = \dfrac{d}{dt}\langle \cos t, \sin t, 0 \rangle = \langle -\sin t, \cos t, 0 \rangle$

$\Rightarrow \quad \vec{\mathbf{F}} \bullet \dfrac{d\vec{\mathbf{r}}}{dt} = -\cos t \sin t + \sin t \cos t + 0 = 0$

$\Rightarrow \quad W = \displaystyle\int_0^{2\pi} 0\, dt = \underline{\underline{0}}$

In this case, the force is orthogonal to the direction of motion at all times and no work is done.

Career Avenues

If the initial and terminal points of a curve C are identical and the curve meets itself nowhere else, then the curve is said to be a **simple closed curve**.

Notation:

When C is a simple closed curve, write $\int_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}}$ as $\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}}$.

F is a **conservative vector field** if and only if $\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = 0$ for all simple closed curves C in the domain.

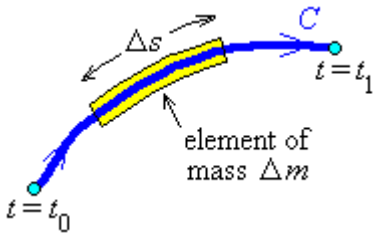
Be careful of where the endpoints are and of the order in which they appear (the orientation of the curve). The identity $\int_{t_0}^{t_1} \vec{\mathbf{F}} \bullet \frac{d\vec{\mathbf{r}}}{dt}\, dt \equiv -\int_{t_1}^{t_0} \vec{\mathbf{F}} \bullet \frac{d\vec{\mathbf{r}}}{dt}\, dt$ leads to the result

$$\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = -\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} \quad \forall \text{ simple closed curves C}$$

Another Application of Line Integrals:

The Mass of a Wire:

Let C be a segment ($t_0 \le t \le t_1$) of wire of line density $\rho(x, y, z)$. Then



$$\Delta m \approx \rho\, x, y, z\ \Delta s$$

$$\Rightarrow \quad m = \int_C \rho\, ds = \int_C \rho \frac{ds}{dt}\, dt = \int_{t_0}^{t_1} \rho \frac{ds}{dt}\, dt$$

First moments about the coordinate planes:

$$\Delta\vec{\mathbf{M}} = \vec{\mathbf{r}}\, \Delta m \approx \rho \vec{\mathbf{r}}\, \Delta s \Rightarrow \vec{\mathbf{M}} = \int_{t_0}^{t_1} \rho \vec{\mathbf{r}} \frac{ds}{dt}\, dt$$

The location $\langle \vec{\mathbf{r}} \rangle$ of the centre of mass of the wire is $\langle \vec{\mathbf{r}} \rangle = \frac{\vec{\mathbf{M}}}{m}$, where

$$\vec{\mathbf{M}} = \int_{t_0}^{t_1} \rho \vec{\mathbf{r}} \frac{ds}{dt}\, dt, \quad m = \int_{t_0}^{t_1} \rho \frac{ds}{dt}\, dt \quad \text{and} \quad \frac{ds}{dt} = \left| \frac{d\vec{\mathbf{r}}}{dt} \right| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}.$$

Career Avenues

Example:

Find the mass and centre of mass of a wire C (described in parametric form by $x = \cos t$, $y = \sin t$, $z = t$, $-\pi \le t \le \pi$) of line density $\rho = z^2$.

Let $c = \cos t$, $s = \sin t$.

[The shape of the wire is one revolution of a helix, aligned along the z axis, centre the origin.]

$$\vec{\mathbf{r}} = \langle c, s, t \rangle \quad \Rightarrow \quad \frac{d\vec{\mathbf{r}}}{dt} = \langle -s, c, 1 \rangle$$

$$\Rightarrow \quad \frac{ds}{dt} = \sqrt{-s^{\,2} + c^2 + 1^2} = \sqrt{2}$$

$$\rho = z^2 = t^2$$

$$\Rightarrow \quad m = \int_C \rho \, ds = \int_{-\pi}^{\pi} \rho \frac{ds}{dt} \, dt = \sqrt{2} \int_{-\pi}^{\pi} t^2 \, dt = \sqrt{2} \left[\frac{t^3}{3} \right]_{-\pi}^{\pi}$$

$$\Rightarrow \quad m = \underline{\underline{\frac{2}{3} \sqrt{2} \, \pi^3}}$$

$$\vec{\mathbf{M}} = \int_{t_0}^{t_1} \rho \vec{\mathbf{r}} \frac{ds}{dt} \, dt = \sqrt{2} \int_{-\pi}^{\pi} \left\langle t^2 c, t^2 s, t^3 \right\rangle dt$$

x component:-

D	I
t^2	c
$2t$	s
2	$-c$
0	$-s$

Integration by parts:-

$$\int t^2 c \, dt = \left[t^2 - 2 \; s + 2tc \right]$$

$$\Rightarrow \quad \int_{-\pi}^{\pi} t^2 c \, dt = \left[t^2 - 2 \; s + 2tc \right]_{-\pi}^{\pi}$$

$$= \quad 0 - 2\pi \quad - \quad 0 + 2\pi \quad = -4\pi$$

y component:-

For all integrable functions f (t) and for all constants a note that

$$\int_{-a}^{a} f \; t \; dt = \begin{cases} 0 & \text{if } f \;\; t \text{ is an ODD function} \\ 2 \int_0^a f \; t \; dt & \text{if } f \;\; t \text{ is an EVEN function} \end{cases}$$

$t^2 \sin t$ is an odd function

$$\Rightarrow \quad \int_{-\pi}^{\pi} t^2 s \, dt = 0$$

z component:-

t^3 is also an odd function

$$\Rightarrow \quad \int_{-\pi}^{\pi} t^3 \, dt = 0$$

Career Avenues

Therefore $\vec{\mathbf{M}} = -4\pi\sqrt{2}\,\hat{\mathbf{i}}$

$\langle\vec{\mathbf{r}}\rangle = \dfrac{\vec{\mathbf{M}}}{m} = \dfrac{3}{2\pi^3\sqrt{2}} - 4\pi\sqrt{2}\,\hat{\mathbf{i}} = -\dfrac{6}{\pi^2}\,\hat{\mathbf{i}}$

The centre of mass is therefore at $\underline{\underline{\left(-\dfrac{6}{\pi^2}, 0, 0\right)}}$

Conservative Vector Fields

If the vector field $\mathbf{F}\ x,y,z = \langle F_1\ x,y,z\ , F_2\ x,y,z\ , F_3\ x,y,z\ \rangle$ has components with continuous first partial derivatives in a simply-connected region on D, then the following five statements are equivalent (i.e., they are either all true or all false)

$\mathbf{F}\ x,y,z$ is conservative in D.

1. $\mathbf{F}\ x,y,z$ is a gradient field in D. (i.e., $\mathbf{F}\ x,y = \nabla f\ x,y$).
2. $\int_C \mathbf{F}\ x,y,z \cdot d\mathbf{r}$ is independent of path in D.
3. $\int_C \mathbf{F}\ x,y,z \cdot d\mathbf{r} = 0$ for every piecewise-smooth closed curve C lying in D.
4. $\mathbf{F}\ x,y,z$ is irrotational. (i.e., $\nabla \times \mathbf{F} = \mathbf{0}$).

3.7.2 Surface Integral:-

Any integral which is to be evaluated over a surface is called surface integral and is denoted by

$\iint$ f.uds where u is the unit vector normal to the surface S.

Evaluation of surface integrals:-

To evaluate surface integral, we have to project the surface to the rectangular co-ordinate planes (XY or XZ or YZ).

Case I:-

If the orthogonal projection of the function on the XY-plane, then

surface integral = $\iint$ f.uds where

u = k & ds = dxdy

1u. kl

Case 2:-

If the orthogonal projection of the function on the YZ plane, then

Surface integral = $\iint$ f.uds where

u = k & ds = dydz

Case 3:-

If the orthogonal projection of the function on the XZ plane, then

surface integral – = $\iint$ f.uds where

Career Avenues

ds = dxdz

Example:- Evaluate ∬ f.uds where f = xzi + ycj + z²k and S is the upper half of surface of the sphere x² + y² + z² = 1
Let ϕ = x² + y² + z² - 1.
u = vϕ = 2xi + 2yj + 2zk = xi + yj + zk
1vϕ1 √2
To project the given unit sphere to the XY plane, then
∫s∫ f.uds = ∫s∫ z.dxdy - ∫s∫ dxdy
1u. kl
= Area of the unit circle
= π.1² = π

3.7.3 Volume Integral:

Any integral which is to be evaluated over a volume is called a volume integral, it is denoted by ∫v∭ fdv. We can subdivide the region region V into small cuboids by drawing lines parallel to the 3 co-ordinate axes. Then

dv = dxdydz

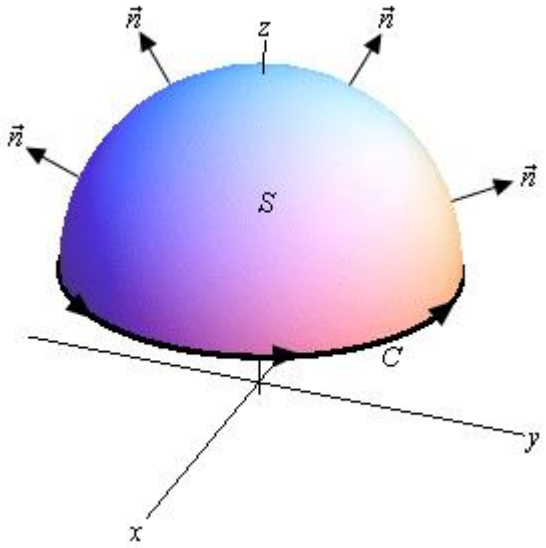
Example:
If f = 2xzi – xj + y²k and V is the region bounded by the surfaces.
X = 0, y = 0, y = 6, z = x² and z = w. Then
Evaluate ∫v∭ fdv.
∫v∭ fdv. =² ∫o⁶∫ o⁴∫ o fdv. (2xzi – xj + y²k) dxdydz
= 192i – 48j + 5761c

3.8 Stokes Theorem:

In this section we are going to take a look at a theorem that is a higher dimensional version of Green's Theorem. In Green's Theorem we related a line integral to a double integral over some region. In this section we are going to relate a line integral to a surface integral. However, before we give the theorem we first need to define the curve that we're going to use in the line integral.

Let's start off with the following surface with the indicated orientation.

Career Avenues



Around the edge of this surface we have a curve C. This curve is called the **boundary curve**. The orientation of the surface S will induce the **positive orientation of C**. To get the positive orientation of C think of yourself as walking along the curve. While you are walking along the curve if your head is pointing in the same direction as the unit normal vectors while the surface is on the left then you are walking in the positive direction on C.

Now that we have this curve definition out of the way we can give Stokes' Theorem.

Theorem:

Let S be an oriented smooth surface that is bounded by a simple, closed, smooth boundary curve C with positive orientation. Also let be a vector field then,

$$\int_C \vec{F} \bullet d\vec{r} = \iint_S curl\, \vec{F} \bullet d\vec{S}$$

In this theorem note that the surface S can actually be any surface so long as its boundary curve is given by C. This is something that can be used to our advantage to simplify the surface integral on occasion.

Example:
Use Stokes' Theorem to evaluater

$$\iint_S curl\, \vec{F} \bullet d\vec{S}$$

where $\vec{F} = z^2\vec{i} - 3xy\,\vec{j} + x^3 y^3 \vec{k}$

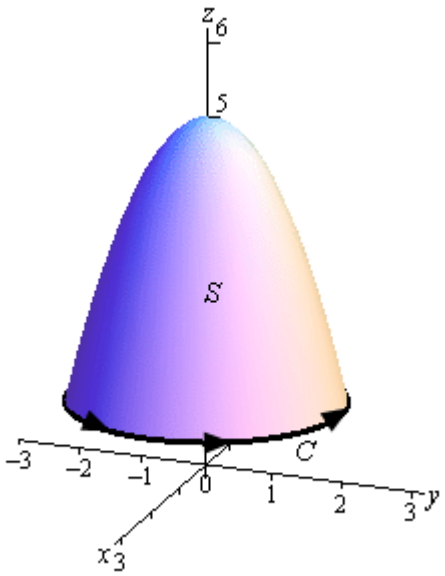
and S is the part of $z = 5 - x^2 - y^2$ above the p lane z = 1

Assume that S is oriented upwards.

Solution:-

Let's start this off with a sketch of the surface

Career Avenues



In this case the boundary curve C will be where the surface intersects the plane z = 1 and so will be the curve

$1 = 5 - x^2 = y^2$

$x^2 + y^2 = 4 \qquad at\ z = 1$

So, the boundary curve will be the circle of radius 2 that is in the plane z = 1. The parameterization of this curve is,

$\vec{r}\ t = 2\cos t\,\vec{i} + 2\sin t\,\vec{j} + \vec{k}, 0 \le t \le 2\pi$

So, it looks like we need a couple of quantities before we do this integral. Let's first get the vector field evaluated on the curve. Remember that this is simply plugging the components of the parameterization into the vector field.

$\vec{F}\ \vec{r}\ t\ = 1^{\ 2}\vec{i} - 3\ 2\cos t\ \ 2\sin t\ \vec{j} + 2\cos t^{\ 3}\ \ 2\sin t^{\ 3}$

$= \vec{i} - 12\cos t \sin \vec{j} + 64\ \cos^3 t \sin^3 t\,\vec{k}$

Next, we need the derivative of the parameterization and the dot product of this and the vector field.

$\vec{r}\ t = -2\sin t\,\vec{i} + 2\cos t\,\vec{j}$

$\vec{F}\ \vec{r}\ t\ \bullet \vec{r}\ t = -2\sin t - 24\sin t\cos^2 t$

$$\iint_S curl\,\vec{F} \bullet d\vec{S} = \int_0^{2\pi} -2\sin t - 24\sin t\cos 2t\,dt$$

$$= 2\cos t + 8\cos^3 t\ \Big|_0^{2\pi}$$

$$= 0$$

Example:

Use Stokes' Theorem to evaluate

$\int_C \vec{F} \bullet d\vec{r}$ Where, $\vec{F} = z^2\,\vec{j} + y^2\,\vec{j} + x\vec{k}$

and C is the triangle with vertices $1,0,0 \quad 0,1,0 \quad 0,0,1$ with counter-clockwise rotation.

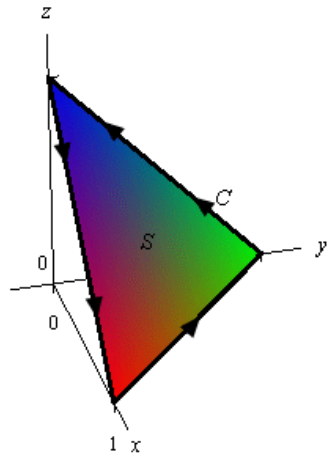
Solution:-

We are going to need the curl of the vector field eventually so let's get that out of the way first.

Career Avenues

$$curl\,\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ z^2 & y^2 & x \end{vmatrix} = 2z\,\vec{j} - \vec{j} = \left\langle 0, 2z-1, 0 \right\rangle$$

Now, all we have is the boundary curve for the surface that we'll need to use in the surface integral. However, as noted above all we need is any surface that has this as its boundary curve. So, let's use the following plane with upwards orientation for the surface.



Since the plane is oriented upwards this induces the positive direction on C as shown. The equation of this plane is,

$$x + y + z = 1 \qquad \Rightarrow \qquad z = g\left(x,y\right) = 1 - x - y$$

$$\begin{aligned} \int_C \vec{F} \bullet d\vec{r} &= \iint_S curl\,\vec{F} \bullet d\vec{S} \\ &= \iint_S \left\langle 0, 2z-1, 0 \right\rangle \bullet d\vec{S} \\ &= \iint_D \left\langle 0, 2z-1, 0 \right\rangle \bullet \frac{\nabla f}{\left\| \nabla f \right\|} \left\| \nabla f \right\| dA \end{aligned}$$

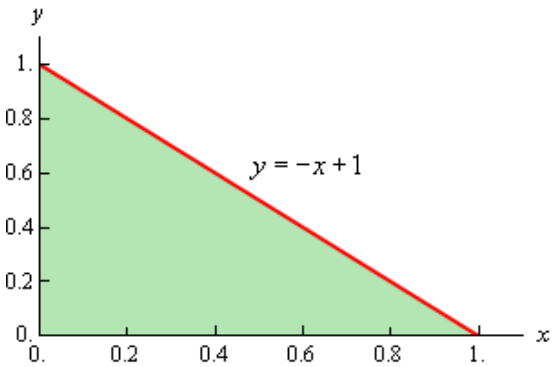
Okay, we now need to find a couple of quantities. First let's get the gradient. Recall that this comes from the function of the surface.

$$f\left(x,y,z\right) = z - g\left(x,y\right) = z - 1 + x + y$$

$$\nabla f = \vec{i} + \vec{j} + \vec{k}$$

Note as well that this also points upwards and so we have the correct direction.

Now, D is the region in the xy-plane shown below.



We get the equation of the line by plugging in $z = 0$ into the equation of the plane. So based on this the ranges that define D are,

$$0 \le x \le 1 \qquad 0 \le y \le -x + 1$$

Career Avenues

$$\int_0 \vec{F} \bullet d\vec{r} = \iint_D \left(2z-1\right)\vec{j} \bullet \left(\vec{i}+\vec{j}+\vec{k}\right) dA$$

$$= \int_0^1 \int_0^{-z+1} 2\left(1-x-y\right) - 1 dy\, dx$$

Don't forget to plug in for z since we are doing the surface integral on the plane. Finishing this out gives.

$$\int_C \vec{F} \bullet d\vec{r} = \int_0^1 \int_0^{-z+1} 1-2x-2y\, dy\, dx$$

$$= \int_0^1 \left. y-2xy-y^2 \right|_0^{-x+1} dx$$

$$= \int_0^1 x^2 - x\, dx$$

$$= \left(\frac{1}{3}x^3 - \frac{1}{2}x^2\right)\Big|_0^1$$

$$= -\frac{1}{6}$$

Career Avenues

Example 3:-Verify Stokes's Theorem to evaluate the line integral $\int_C z^2dx + x^2dy + y^2dz$ where S is the part of x + y + z = 2 in the first octant
and C is the boundary of the triangle.

Solution:-

$\vec{F}(x,y,z) = z^2dx + x^2dy + y^2dz$

$$Curl\,\vec{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & x^2 & y^2 \end{vmatrix} = i(2y-0) - j(0-2z) + k(2x-0) = 2yi + 2zj + 2xk$$

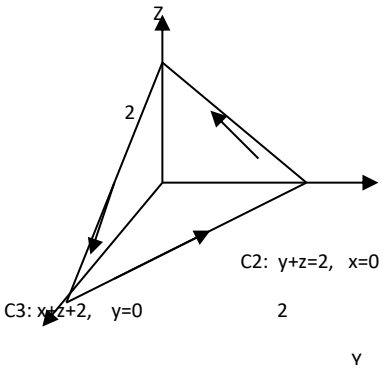
$$z = 2 - x - y \rightarrow \vec{n} = \frac{-(-1)i - (-1)j + 1k}{\sqrt{3}} = \frac{i + j + k}{\sqrt{3}}$$

$$\rightarrow curl\vec{F} \bullet \vec{n} = \ 2yi + 2zj + 2xk \ \bullet \frac{i + j + k}{\sqrt{3}} = \frac{2y + 2z + 2x}{\sqrt{3}} = \frac{2}{\sqrt{3}}\ y + z + x$$

$$dS = \sqrt{1 + f_x^2 + f_y^2}\,dxdy \rightarrow dS = \sqrt{3}dxdy$$

$$\int_C z^2dx + x^2dy + y^2dz = \iint_D \frac{2}{\sqrt{3}}\ y + z + x\ \sqrt{3}dxdy = 2\int_0^2 \int_0^{2-y} (y + (2 - x - y) + x)dxdy$$

$$= 4\int_0^2 \int_0^{2-y} dxdy = 4\int_0^2 (2-y)dy = 4\left[2y - \frac{y^2}{2}\right]\Bigg|_0^2 = 4\ 4 - 2\ = 8$$



$$C1: \int_{C1} z^2dx + x^2dy + y^2dz = \int_0^2 0 + (2-y)^2dy + 0 = -\frac{(2-y)^3}{3}\Bigg|_0^2 = \frac{8}{3}$$

$$C2: \int_{C2} z^2dx + x^2dy + y^2dz = \int_0^2 0 + 0dy + \ 2 - z\ ^2\,dz = -\frac{(2-z)^3}{3}\Bigg|_0^2 = \frac{8}{3}$$

$$C3: \int_{C3} z^2dx + x^2dy + y^2dz = \int_0^2 \ 2 - x\ ^2\,dx + 0 + 0 = -\frac{(2-x)^3}{3}\Bigg|_0^2 = \frac{8}{3}$$

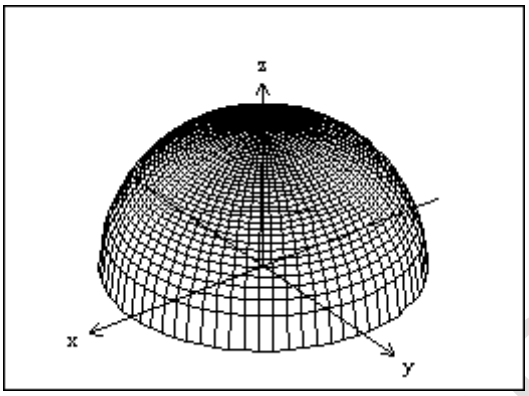
$$\int_C z^2dx + x^2dy + y^2dz = \frac{8}{3} + \frac{8}{3} + \frac{8}{3} = 8$$

Career Avenues

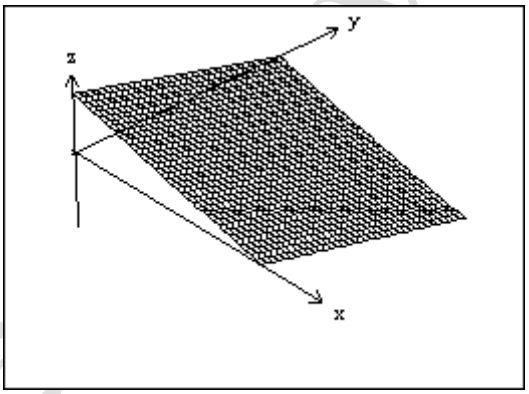
Practice Exercise

1. Verify Stoke's theorem for the vector field
F = yz **i** + xz **j** + xy **k** over the surface S: z = x^2 + y^2 and z<=1.

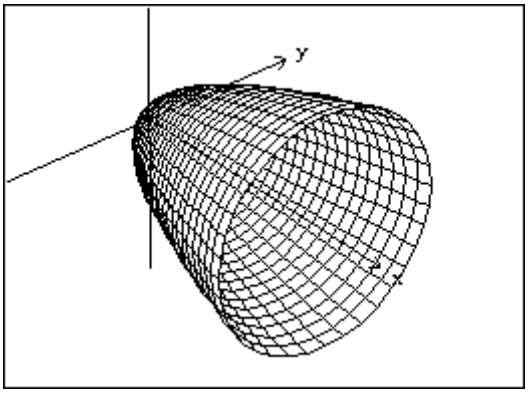
2. Verify Stokes's theorem by computing both integrals for S is the portion of $z = \sqrt{1 - x^2 - y^2}$ above the x-y plane and $\mathbf{F} = \langle 2x, z^2 - x, xz^2 \rangle$.



3. Use Stokes's Theorem to compute $\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} dS$ where S is the portion of the tetrahedron bounded by $x + y + 4z = 8$ and the coordinate planes with z > 0, **n** upward, $\mathbf{F} = \langle y^2, y + 2x, z^2 \rangle$.



4. Use Stokes's Theorem to compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is the boundary of the portion of the paraboloid $x = y^2 + z^2$ with x ≤ 4, **n** to the back, $\mathbf{F} = \langle yz, y - 4, 2xy \rangle$.



Career Avenues

5. Let S be the surface of a cut hollow sphere whose base is the unit circle lying in the xy-plane. Let **F** = y **i** - x **j** + e^{xz} **k.** Calculate the flux of the curl of **F** across S.

6. Calculate the circulation of the field **F** = x^2 **i** + 2x **j** + z^2 **k** around the curve C: the ellipse $4x^2 + y^2 = 4$ in the xy-plane, counter-clockwise when viewed from above.

7. Calculate the circulation of the field **F** = y **i** + xz **j** + x^2 **k** around the curve C: the boundary of the triangle cut from the plane x + y + z = 1 by the first octant, counter-clockwise when viewed from above.

3.9 Divergence Theorem:

In this section we are going to relate surface integrals to triple integrals. We will do this with the Divergence Theorem.

Let E be a simple solid region and S is the boundary surface of E with positive orientation. Let $\vec{F}$ be a vector field whose components have continuous first order partial derivatives. Then,

$$\iint_S \vec{F} \bullet d\vec{S} = \iiint_{E\,dV} div\ \vec{F}$$

Let's see an example of how to use this theorem.

Example:

Use the divergence theorem to evaluate $\iint_S \vec{F} \bullet d\vec{S}$ where

$\vec{F} = xy\vec{i} - \frac{1}{2}y^2\vec{j} + z\vec{k}$ and the surface consists of the three surfaces,

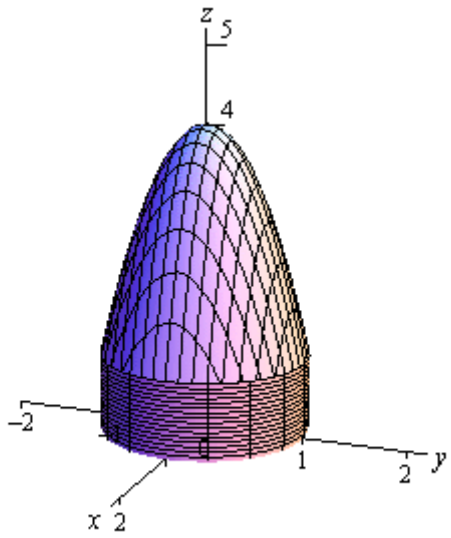
$z = 4 - 3x^2 - 3y^2, 1 \le z \le 4$ on the top, $x^2 + y^2 = 1$, $0 \le z \le 1$ on the sides and

$z = 0$ on the bottom.

Solution:-

Let's start this off with a sketch of the surface.

Career Avenues



The region E for the triple integral is then the region enclosed by these surfaces.
Note that cylindrical coordinates would be a perfect coordinate system for this region.
If we do that here are the limits for the ranges.

$0 \le z \le 4 - 3r^2$

$0 \le r \le 1$

$0 \le \theta \le 2\pi$

We'll also need the divergence of the vector field so let's get that.

$$div\,\vec{F} = y - y + 1 = 1$$

The integral is then,

$$\iint_S \vec{F} \bullet d\vec{S} = \iiint_E div\,\vec{F}\, dV$$

$$= \int_0^{2\pi} \int_0^1 \int_0^{4-3r^2} r\, dz\, dr\, d\theta$$

$$= \int_0^{2\pi} \int_0^1 4r - 3r^3\, dr\, d\theta$$

$$= \int_0^{2\pi} \left(2r^2 - \frac{3}{4} r^4 \right) \Big|_0^1 d\theta$$

$$= \int_0^{2\pi} \frac{5}{4} d\theta$$

$$= \frac{5}{2}\pi$$

3.10 Green's Theorem:

A curve C on $\mathbb{R}^2$ (defined in parametric form by $\vec{\mathbf{r}}\ t\ = x\ t\ \hat{\mathbf{i}} + y\ t\ \hat{\mathbf{j}}$, $a \le t \le b$) is **closed**

If f (x(a), y(a)) = (x(b), y(b)) .

The curve is **simple** if

$\vec{\mathbf{r}}\ t_1 \neq \vec{\mathbf{r}}\ t_2$ for all t_1, t_2 such that $a < t_1 < t_2 < b$;

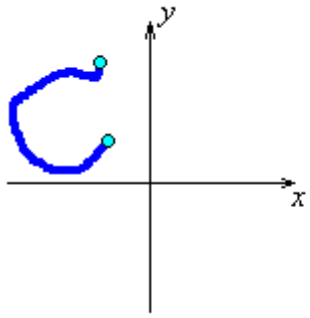
Career Avenues

(that is, the curve neither touches nor intersects itself, except possibly at the end points).

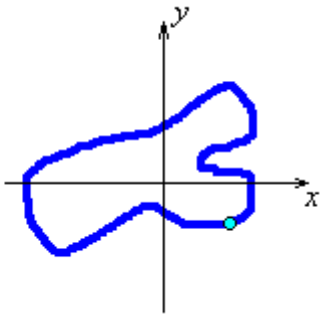
For Example:

Two simple curves:-

Open

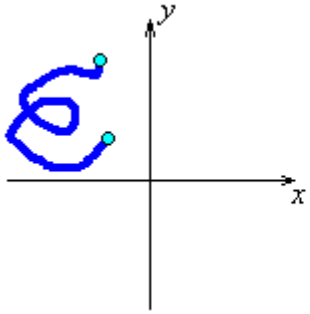


closed

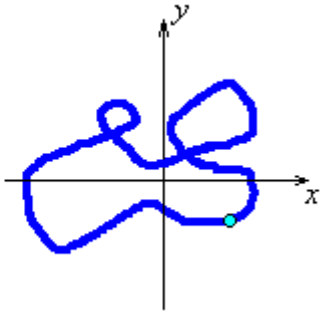


Two non-simple curves:-

open



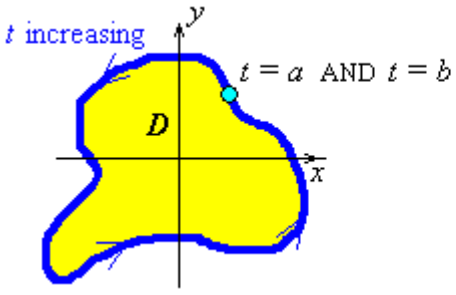
closed



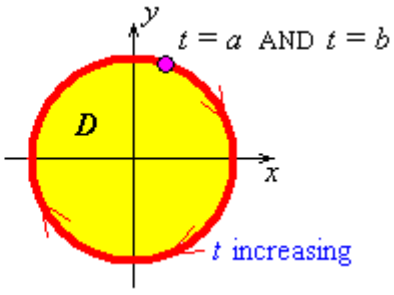
Orientation of closed curves:-

A closed curve C has a positive orientation iff a point **r**(t) moves around C in an anticlockwise sense as the value of the parameter t increases.

For Example:



Positive orientation



Negative orientation

Career Avenues

Let D be the finite region of $\mathbb{R}^2$ bounded by C. When a particle moves along a curve with positive orientation, D is always to the left of the particle.

For a simple closed curve C enclosing a finite region D of $\mathbb{R}^2$ and for any vector function $\vec{\mathbf{F}} = \langle f_1, f_2 \rangle$ that is differentiable everywhere on C and everywhere in D,

Green's theorem is valid:

$$\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \iint_D \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dA$$

The region D is entirely in the xy-plane, so that the unit normal vector everywhere on D is **k**. Let the differential vector **dA** = dA **k** , then Green's theorem can also be written as

$$\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \iint_D \vec{\nabla} \times \vec{\mathbf{F}} \bullet \hat{\mathbf{k}}\, dA = \iint_D \text{curl } \vec{\mathbf{F}} \bullet d\vec{\mathbf{A}}$$

$$\vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \langle f_1, f_2 \rangle \bullet \langle dx, dy \rangle \quad \Rightarrow \quad \oint_C f_1\, dx + f_2\, dy = \iint_D \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dA$$

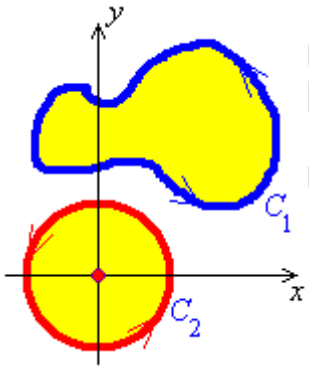
and

$$\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = \det \begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ f_1 & f_2 \end{bmatrix} = \det \begin{bmatrix} \vec{\nabla}^{\mathrm{T}} \\ \vec{\mathbf{F}}^{\mathrm{T}} \end{bmatrix} = \text{ z component of } \vec{\nabla} \times \vec{\mathbf{F}}$$

Green's theorem is valid if there are no singularities in D.

For Example:

$\vec{\mathbf{F}} = \left\langle \frac{x}{r}, 0 \right\rangle$:



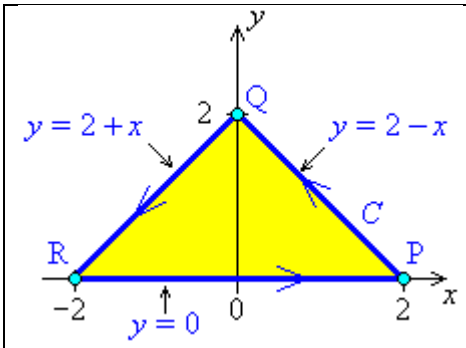
Green's theorem is valid for curve C_1 but not for curve C_2.

There is a singularity at the origin, which curve C_2 encloses.

Example:

For $\vec{\mathbf{F}} = \langle x + y, x - y \rangle$ and C as shown, evaluate $\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}}$.

Career Avenues



$$\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \int_{PQ} \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} + \int_{QR} \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} + \int_{RP} \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}}$$

$\vec{\mathbf{F}} = \langle x+y, x-y \rangle$

Everywhere on the line segment from P to Q, y = 2 – x (and the parameter t is just x)

$\Rightarrow \quad \vec{\mathbf{r}} = \langle x, 2-x \rangle \quad \Rightarrow \quad \dfrac{d\vec{\mathbf{r}}}{dx} = \langle 1, -1 \rangle$ and $\vec{\mathbf{F}} = \langle 2, 2x-2 \rangle$

$$\Rightarrow \int_{PQ} \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \int_2^0 2 - 2x - 2 \; dx = \int_2^0 4 - 2x \; dx = \left[4x - x^2 \right]_2^0$$

$= (0 - 0) - (8 - 4) = -4$

Everywhere on the line segment from Q to R, y = 2 + x

$\Rightarrow \quad \vec{\mathbf{r}} = \langle x, 2+x \rangle \quad \Rightarrow \quad \dfrac{d\vec{\mathbf{r}}}{dx} = \langle 1, 1 \rangle$ and $\vec{\mathbf{F}} = \langle 2x+2, -2 \rangle$

$$\Rightarrow \int_{QR} \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \int_0^{-2} 2x + 2 - 2 \; dx = \int_0^{-2} 2x \; dx = \left[x^2 \right]_0^{-2}$$

$= 4 - 0 = 4$

Everywhere on the line segment from R to P, y = 0

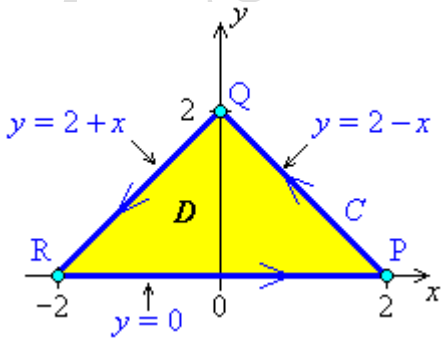
$\Rightarrow \quad \vec{\mathbf{r}} = \langle x, 0 \rangle \quad \Rightarrow \quad \dfrac{d\vec{\mathbf{r}}}{dx} = \langle 1, 0 \rangle$ and $\vec{\mathbf{F}} = \langle x, x \rangle$

$$\Rightarrow \int_{RP} \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \int_{-2}^2 x + 0 \; dx = \int_{-2}^2 x \; dx = \left[\frac{x^2}{2} \right]_{-2}^2$$

$= 2 - 2 = 0$

$$\Rightarrow \oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = -4 + 4 + 0 = \underline{\underline{0}}$$

OR use Green's theorem!



Career Avenues

$$\det\begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ f_1 & f_2 \end{bmatrix} = \frac{\partial}{\partial x}\, x - y \; - \frac{\partial}{\partial y}\, x + y \; = 1 - 1 = 0$$

everywhere on D

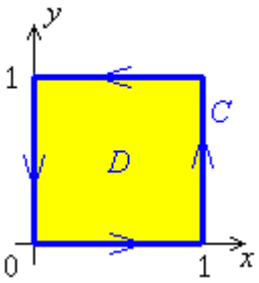
$$\Rightarrow \iint_D \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dA = \iint_D 0 \, dA = 0$$

By Green's theorem it then follows that

$$\oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \underline{\underline{0}}$$

Example:

Find the work done by the force $\vec{\mathbf{F}} = \langle xy, y^2 \rangle$ in one circuit of the unit square.



By Green's theorem,

$$W = \oint_C \vec{\mathbf{F}} \bullet d\vec{\mathbf{r}} = \iint_D \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dA$$

$$\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} = \frac{\partial}{\partial x}\, y^2 \; - \frac{\partial}{\partial y}\, xy \; = 0 - x$$

The region of integration is the square $0 < x < 1, \; 0 < y < 1$

$$\Rightarrow \; W = \iint_D -x \, dA = \int_0^1 \int_0^1 -x \; dy \, dx$$

$$= -\int_0^1 x \left(\int_0^1 1 \, dy \right) dx = -\int_0^1 x \; y \, \Big|_0^1 \, dx = -\int_0^1 x \; 1 - 0 \; dx$$

$$= -\left[\frac{x^2}{2} \right]_0^1 = -\frac{1}{2} + 0 = -\frac{1}{2}$$

Therefore

$$W = \underline{\underline{-\frac{1}{2}}}$$

The alternative method (using line integration instead of Green's theorem) would involve **four** line integrals, each with different integrands.

Career Avenues

SET- 7

1. The directional derivative of $f\ x,y\ = 2x^2 + 3y^2 + z^2$ at point $P\ 2,1,3$ in the direction of the vector $a = \vec{i} - 2\vec{k}$ *is*

a) $4/\sqrt{5}$

b) $-4/\sqrt{5}$

c) $\sqrt{5}/4$

d) $-\sqrt{5}/4$

2. The derivative of $f\ x,y$ *at point* $1,2$ *in the direction of vector* $i + j$ *is* $2\sqrt{2}$ and in the direction of the *ector* $-2j$ *is* -3 . Then the derivat $f\ x,y$ in direction $-i - 2j$ *is*

a) $2\sqrt{2} + 3\sqrt{2}$

b) $-7/\sqrt{5}$

c) $-2\sqrt{2} - 3\sqrt{2}$

d) $1/\sqrt{2}$

3. The expression *curl grad* f . where f *is a scalar function is*

a) Equal to $\nabla^2 f$.

b) Equal to *div grad* f .

c) A scalar of zero magnitude

d) A vector of zero magnitude

4. For a function $\varnothing = ax^2 y - y^3$ to represent the velocity potential of an ideal fliud, $\nabla^2 \varnothing$ should be eqaul to zero. In that case, the value of 'a' has to be

a) -1

b) 1

c) 3

d) -3

5. Given a vector field $\bar{F}$, the divergence theorem states that

a) $\int_s \bar{F}.\overline{ds} = \int_v \nabla.\bar{F}dv$

b) $\int_s \bar{F}.\overline{ds} = \int_v \bar{\nabla}.\bar{F}dv$

c) $\int_s \bar{F} \times \overline{ds} = \int_v \nabla.\bar{F}dv$

d) $\int_s \bar{F} \times \overline{ds} = \int_v \bar{\nabla}.\bar{F}dv$

Career Avenues

6. The directional derivative of the following function $f\ x$ *at* $1,2$ *in the direction of* $4i+3j$ *is* …. Where $f\ x,y = x^2+y^2$

a) $4/5$
b) 4
c) $2/5$
d) 1

7. Which one of the following is NOT associated with vector calculus?
a) Stoke's theorem
b) Guass Divergence theorem
c) Green's theorem
d) Kennedy's theorem

8. For the scalar field $u=\frac{x^2}{2}+\frac{y^2}{3}$, the magnitude of the gradient at the point (1,3) is

a) $\sqrt{13/9}$
b) $\sqrt{9/2}$
c) $\sqrt{5}$
d) $9/2$

9. The area of triangle formed by the tips of vectors $\bar{a},\bar{b}$ *and* $\bar{c}$ *is*

a) $\frac{1}{2}\ \bar{a}-\bar{b}\ .\ \bar{a}-\bar{c}$

b) $\frac{1}{2}|\ \bar{a}-\bar{b}\ \times\ \bar{a}-\bar{c}\ |$

c) $\frac{1}{2}\left|\bar{a}\times\bar{b}\times\bar{c}\right|$

d) $\frac{1}{2}\ \bar{a}\times\bar{b}\ .\bar{c}\ |$

10. The angle (in degrees) between tw planar vector $\bar{a}=\frac{\sqrt{3}}{2}i+\frac{1}{2}j$ *and* $\bar{b}=\frac{-\sqrt{3}}{2}i+\frac{1}{2}j$ *is*

a) 30
b) 60
c) 90
d) 120

11. For a scalar function $f\ x,y,z\ = x^2+3y^2+2z^2$ *, the gradient at the point* $P\ 1,2,-1$ *is*

a) $2\bar{i}+6\bar{j}+4\bar{k}$
b) $2\bar{i}+12\bar{j}-4\bar{k}$
c) $2\bar{i}+12\bar{j}+4\bar{k}$
d) $\sqrt{56}$

Career Avenues

12. For a scalar function $f(x,y,z) = x^2 + 3y^2 + 2z^2$, The directional derivative at the point P (1,2,-1) in the direction of a vector $\bar{i} - \bar{j} + 2\bar{k}$ is

a) -18
b) $-3\sqrt{6}$
c) $3\sqrt{6}$
d) 18

13. The divergence of the vector field $(x-y)\,\bar{i} + (y-x)\,\bar{j} + (x+y+z)\,\bar{k}$ *is*

a) 0
b) 1
c) 2
d) 3

14. A sphere of unit radius is centered at the origin. The unit normal at a point (x,y,z) on the surface of the sphere is the vector.

a) (x,y,z)
b) $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$
c) $\left(\frac{x}{\sqrt{3}},\frac{y}{\sqrt{3}},\frac{z}{\sqrt{3}}\right)$
d) $\left(\frac{x}{\sqrt{2}},\frac{y}{\sqrt{2}},\frac{z}{\sqrt{2}}\right)$

15. The divergence of the vector field $3xz\bar{i} + 2xy\bar{j} - yz^2\bar{k}$ at a point (1,1,1) is

a) 7
b) 4
c) 3
d) 0

16. The line integral of the vector function $\bar{F} = 2x\bar{i} + x^2\bar{j}$ along the axis from $x = 1$ to $x = 2$ *is*

a) 0
b) 2.33
c) 3
d) 5.33

17. If $T(x,y,z) = x^2 + y^2 + 2z^2$ *defines teh temperature at any location* (x,y,z) *then* the magnitude of the temperature gradient at point $P(1,1,1)$ is

a) $2\sqrt{6}$
b) 4
c) 24
d) $\sqrt{6}$

Career Avenues

18. If $\bar{a}$ *and* $\bar{b}$ *are two* arbitary vectors with magnitudes a nd b respectively, $\left|\bar{a}\times\bar{b}\right|^2$ will be equal to

a) $a^2b^2 - \bar{a}.\bar{b}^{\,2}$

b) $ab - \bar{a}\bar{b}$

c) $a^2b^2 + \bar{a}.\bar{b}^{\,2}$

d) $ab + \bar{a}\bar{b}$

19. For the spherical suface $x^2 + y^2 + z^2 = 1$, the unit outward normal vector at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ *is given by*

a) $\frac{1}{\sqrt{2}}\bar{i} + \frac{1}{\sqrt{2}}\bar{j}$

b) $\frac{1}{\sqrt{2}}\bar{i} - \frac{1}{\sqrt{2}}\bar{j}$

c) $\bar{k}$

d) $\frac{1}{\sqrt{3}}\bar{i} + \frac{1}{\sqrt{3}}\bar{j} + \frac{1}{\sqrt{3}}\bar{k}$

20. The two vectors $\left[1,1,1 \text{ and } 1, a, a^2\right]$ where $a = \frac{-1}{2} + j\frac{\sqrt{3}}{2}$ are

a) Orthogonal
b) Orthonormal
c) Parallel
d) Collinear

21. Stock's theorem connects
a) A line integral and a surface integral
b) A surface integral and a volume integral
c) A line integral and a volume integral
d) Gradient of a function and its surface integral

22. The line integral of the vector function $\bar{F} = 2x\hat{i} + x^2\,\hat{j}$ *along the* $x-$*axis from* $x = 1$ *to* $x = 2$ *is*

a) 0
b) 2.33
c) 3
d) 5.33

Career Avenues

Solution:

Set: Vector Calculus

1. Ans: b)

$f\ x,y = 2x^2 + 3y^2 + z^2 \quad P\ 2,1,3$

$and \qquad \bar{a} = \bar{i} - 2\bar{k})$

$Directional\ derivative = \nabla f_{\ P} \bullet \frac{\bar{a}}{|\bar{a}|} = \frac{-4}{\sqrt{5}}$

2. Ans: b)

Directional derivative of f in the direction

$of\ i + j = 2\sqrt{2}$

$\Rightarrow \nabla f . \frac{\bar{a}}{|\bar{a}|} = 2\sqrt{2} \Rightarrow \left(\frac{\partial f}{\partial x}\bar{i} + \frac{\partial f}{\partial y}\bar{j} + \frac{\partial f}{\partial z}\bar{k} \right)$

$\frac{\bar{i} + \bar{j}}{\sqrt{2}} = 2\sqrt{2} \Rightarrow \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 4 \ldots 1$

similarly

$\nabla f . \frac{\bar{b}}{|\bar{b}|} = -3 \Rightarrow \left(\frac{\partial f}{\partial x}\bar{i} + \frac{\partial f}{\partial y}\bar{j} + \frac{\partial f}{\partial z}\bar{k} \right)$

$\frac{-2\bar{j}}{2} = -3 \Rightarrow -2\frac{\partial f}{\partial y} = -6 \Rightarrow \frac{\partial f}{\partial y} = 3 \ldots . 2$

$sub.\ 2\ in\ 1\ , we\ obtain, \frac{\partial f}{dx} = 1$

$\therefore$

$\nabla f . \frac{\bar{c}}{|\bar{c}|} = \left(\frac{\partial f}{\partial x}\bar{i} + \frac{\partial f}{\partial y}\bar{j} + \frac{\partial f}{\partial z}\bar{k} \right) . \frac{(-\bar{i} - 2\tilde{j})}{\sqrt{5}} = \frac{-7}{\sqrt{5}}$

3. Ans: d)

$Curl\ \ grad\ f\ = 0$

4. Ans: c)

$\phi = ax^2\ y - y^3 = \nabla^2 \phi$

$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$

$> 2ay - 6y = 0 \Rightarrow a = 3$

5. Ans: a)

As per Guass Divergence Theorem $\int_S \bar{F} . \overline{ds} = \int_V \nabla . \bar{F} dv$

Career Avenues

6. Ans: b)

$f\ x,y = x^2+y^2, \bar{a} = 4i+3j \qquad 1,2$

Directional derivative =

$\nabla f . \frac{\bar{a}}{|\bar{a}|} = 2i+4j\ . \frac{4i+3j}{5} = 4$

7. Ans: d)

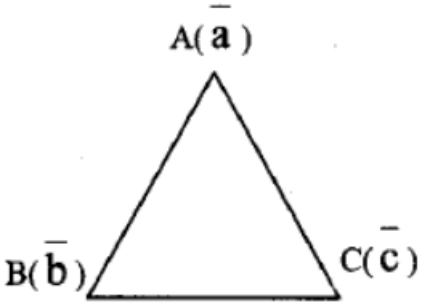
8. Ans: c)

$\nabla u = xi + \frac{2}{3}yj$

$\nabla u_{1,3} = i+2j$

$|\nabla u| = \sqrt{1+4} = \sqrt{5}$

9. Ans: b)



Area of triangle ABC = $\frac{1}{2}\left|\overline{AB}\times\overline{AC}\right|$

$= \frac{1}{2}\left| \bar{b}-\bar{a} \times \bar{c}-\bar{a} \right|$

$= \frac{1}{2}\left| \bar{a}-\bar{b} \times \bar{a}-\bar{c} \right|$

10. Ans: d)

$\bar{a} = \frac{\sqrt{3}}{2}i + \frac{1}{2}j, \quad \bar{b} = -\frac{\sqrt{3}}{2}i + \frac{1}{2}j$

$\cos\theta = \frac{\bar{a}.\bar{b}}{|\bar{a}||\bar{b}|} = \frac{-\frac{3}{4}+\frac{1}{4}}{\sqrt{\frac{3}{4}+\frac{1}{4}}.\sqrt{\frac{3}{4}+\frac{1}{4}}} = -\frac{1}{2}$

$\Rightarrow \theta = 120$

11. Ans:b)

$f\ x,y,z = x^2+3y^2+2z^2\ \ P\ 1,2,-1$

$grad\ f = \nabla f = 2xi+6yj+4z\bar{k}$

$\nabla f_{\ P} = 2i+12\ j-4\bar{k}$

Career Avenues

12. Ans: b)

$f(x,y,z) = x^2 + 3y^2 + 2z^2, \qquad P(1,2,-1)$

$\bar{a} = i - j + 2\bar{k}$

$\nabla f = 2xi + 6yj + 4z\bar{k}$

$\nabla f_P = 2i + 12j - 4\bar{k}$

Directional derivative

$$= \nabla f \cdot \frac{\bar{a}}{|\bar{a}|} = \frac{2 - 12 - 8}{\sqrt{1+1+4}} = -\frac{18}{\sqrt{6}} = -3\sqrt{6}$$

13. Ans: d)

$\bar{V} = (x-y)\,i + (y-x)\,j + (x+y+z)\,\bar{k}$

$div\,\bar{V} = 1 + 1 + 1 = 3$

14. Ans: a)

Equation of sphere is $x^2 + y^2 + z^2 = 1$

Lit $\phi(x,y,z) = x^2 + y^2 + z^2$

Normal to the surface is

$\nabla\phi = 2xi + 2yj + 2z\bar{k}$

Unit normal is $\frac{\nabla\phi}{|\nabla\phi|} = \frac{2(xi + yj + z\bar{k})}{\sqrt{4x^2 + 4y^2 + 4z^2}}$

$= xi + yj + z\bar{k} \left[\because x^2 + y^2 + z^2 = 1\right]$

15. Ans: b)

$\bar{f} = 3xz\,\bar{i} + 2xy\,\bar{j} - yz^2\,\bar{k}$

$div\,\bar{f} = 3z + 2y - 2yz$

$\left[div\,\bar{f}\right]_{1,1,1} = 3 + 2 - 2 = 3$

16. Ans: c)

$\bar{F} = 2x\bar{i} + x^2\,\bar{j}$

$Along\ x-axis,\ y = 0 \Rightarrow dy = 0$

$$\int_c \bar{F}.d\bar{r} = \int_c 2x\,dx + x^2\,dy = \int_1^1 2x\,dx = 3$$

Career Avenues

17. Ans: a)

$\nabla T = 2x\bar{i} + 2y\bar{j} + 4z\bar{k}$

$\nabla T_{P\,1,1,1} = 2\bar{i} + 2\bar{j} + 4\bar{k}$

$|\nabla T| = \sqrt{24} = 2\sqrt{6}$

18. Ans: a)

$|\bar{a} \times \bar{b}|^2 = |\bar{a}|^2 |\bar{b}|^2 \sin^2 \bar{a}, \bar{b}$

$a^2 b^2 \; 1 - \cos^2 \bar{a}, \bar{b}$

$a^2 b^2 \; 1 - \bar{a}.\bar{b}^{\,2} / a^2 b^2$

$= a^2 b^2 - \bar{a}.\bar{b}^{\,2}$

19. Ans: a)

$\phi \; x, y, z = x^2 + y^2 + z^2$

$\nabla\phi = 2 \; x\bar{i} + y\bar{j} + z\bar{k}$

$At\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$

$= 2\left(\frac{1}{\sqrt{2}}\bar{i} + \frac{1}{\sqrt{2}}\bar{j} + 0\right) = \sqrt{2} \; \bar{i} + \bar{j}$

$\bar{N} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{\sqrt{a \; i + j}}{\sqrt{\sqrt{2}^{\,2} + \sqrt{2}^{\,2}}} = \frac{\sqrt{2} \; i + j}{\sqrt{2} \; \sqrt{2}}$

$= \frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$

20. Ans: a)

$p = 1,1,1 \; and \; q = \left[1, a, a^2\right]$ where $a = \frac{-1}{2} + j\frac{\sqrt{3}}{2}$

$\bar{P} = 1,1,1$

$\bar{q} = \left[1, a, a^2\right] where \, a = \frac{-1 + j\sqrt{3}}{2}$

$\bar{p}.\bar{q} = 1 + a + a^2$

$= 1 + \left(\frac{-1 + j\sqrt{3}}{2}\right) + \left(\frac{-1 - j\sqrt{3}}{2}\right) = 0$

$\Rightarrow$ The vectors are orthogonal

21. Ans: a)

$\int_c \bar{F}.d\bar{r} = \iint_a curl \, \bar{F}.d\bar{s}$

Career Avenues

22. Ans: c)

$\overline{F} = 2x\bar{i} + x^2\bar{j}$

$Along\ x-axis, y = 0 \Rightarrow dy = 0$

$$\int_c \overline{F}.d\bar{r} = \int_c 2xdx + x^2\,dy = \int_l^2 2x\,dx = 3$$

Career Avenues